

ELEMENTS OF ENGINEERING

LABORATORY MANUAL

Course Code: ME144

B. Tech 1st Year (CE/IT/EC)

ME – 144 ELEMENTS OF ENGINEERING	
CO1	Understand and interpret various engineering drawing.
CO2	Learn the concepts and be able to draw engineering scale, engineering curve, projection & section of solid, orthographic and isometric drawing.
CO3	Understand the overview of computer aided drafting.
CO4	Understand fundamental principles, theory and applications of mechanical engineering which plays an important role in industries.
CO5	Learn the formation of different types of steam and utilize it for the boiler.
CO6	Understand the basics of internal combustion engine, refrigeration and air conditioning system.
CO7	Understand importance and application of civil engineering.
CO8	Understand the fundamentals of surveying and be able to carry out simple survey exercise.
CO9	Learn about different building components, building planning and design of residential building.

List of Experiment (ME144-EOE)		
Sr. No.	Title	Course Outcomes
1	Orthographic Projections & Isometric View	CO1,CO2
2	Computer Aided Drafting & Modeling	CO3
3	Introduction to Surveying	CO7,CO8
4	Linear Measurement-By Chain Survey	CO7,CO8
5	Construction Site Visit	CO7, CO9
6	Building Components Drawing	CO9

CERTIFICATE

*This is to certify that Mr. /Ms. _____
of _____ Class, Roll No. _____
Exam No. _____ has satisfactorily completed his / her term
work in _____ for
the term ending _____ in 20____ / 20____.*

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
CHANGA – 388 421**

Date :

Sign of the Faculty

Head of the Department

INDEX

Sr. No.	Date	Sheet/Experiment No.	Page No.	Marks/ Grade	Date of Assessm ent	Sign of Faculty
1		Orthographic Projections & Isometric View				
2		Computer Aided Drafting and Modeling				
3		Introduction To Surveying				
4		Linear Measurement - By Chain Survey				
5		Construction Site Visit				
6		Building Components Drawing				

Drawing Equipment's & Materials (for Laboratory work)

- ☐ Mini Drafter.
- ☐ Set squares = 45° & 30°- 60° (Within built French curves and protractor).
- ☐ Instrument Box (Engineering Compass Box).
- ☐ Eraser and Drawing clips (or pins).
- ☐ 0.5 mm clutch pencil (with H & 2H Lead only).
- ☐ Stencils (Stencil 5, 6 & 8 mm - Small and Capital letters).
- ☐ Circle master, Scale and Roller scale.
- ☐ Sketch Books (A3 size), Drawing sheets (A2 size) and Sheet container.

How to begin your drawing?

- ☐ Clean the drawing board and all the drawing instruments using handkerchief.
- ☐ Fix the drawing sheet on the drawing board (table).
- ☐ Fix the mini-drafter in convenient position.
- ☐ Draw borderlines on sheet
- ☐ Spacing of drawing between two problems /view is to be planned before the commencement of the drawing.
- ☐ Print the problem number on the left top and then commence the drawing work.

Important guidelines for students:

- ☐ Always be punctual in time. Latecomer won't be permitted without solid reason.
- ☐ Before starting each sheet, signature of concern batch teacher should be taken on the sheet without fail; else no credit would be given to that practical sheet.
- ☐ Students should bring the drawing sheet ready for the practical. The borderlines and
- ☐ Title block should be drawn on the drawing sheet before coming for the practical.
- ☐ Before starting each sheet in the college, each student will have to ensure that the work in the sketch Book pertaining to that sheet is completed in all respect; else the student will not be allowed to start his work in the sheet.
- ☐ Batch wise problems will be drawn on the sheet in the scheduled practical turn in the drawing hall only.
- ☐ Any data written on the sheets should be in the block (CAPITAL) letters only.
- ☐ All problems of all sheets should be drawn by first angle projection method if not specify.
- ☐ Name and ID No. Should be written on sheet in the title block with the ball pen.

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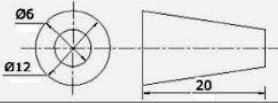
CHANDUBHAI S. PATEL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF MECHANICAL ENGINEERING

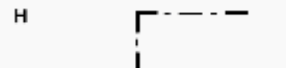
Elements of Engineering (ME144)

F.Y.B.TECH

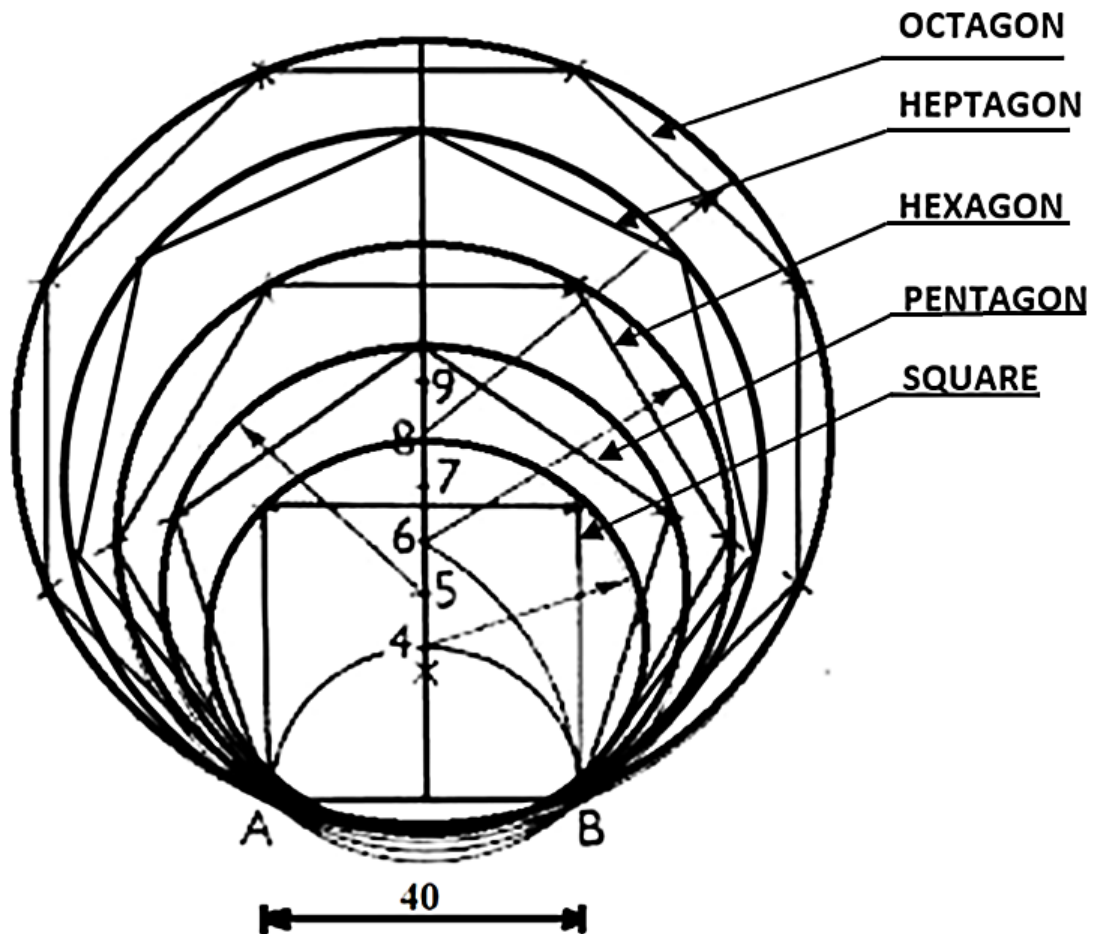
1. TITLE BLOCK:

20		15		15		135	
DATE	SIGN	CHANDUBHAI S. PATEL INSTITUTE OF TECHNOLOGY					
STD		CHANGA-388421					
FAIR		TITLE OF SHEET					
COMP							
NAME							
ID NO.							
F.Y. B.TECH						DRG No. 1/4	
SCALE :		65				70	

2. TYPES OF LINES:

Line	Description	General Applications
A 	Continuous thick	A1 Visible outlines
B 	Continuous thin (straight or curved)	B1 Imaginary lines of intersection B2 Dimension lines B3 Projection lines B4 Leader lines B5 Hatching lines B6 Outlines of revolved sections in place B7 Short centre lines
C 	Continuous thin, free-hand	C1 Limits of partial or interrupted views and sections, if the limit is not a chain thin
D 	Continuous thin (straight) with zigzags	D1 Line (see Fig. 2.5)
E 	Dashed thick	E1 Hidden outlines
G 	Chain thin	G1 Centre lines G2 Lines of symmetry G3 Trajectories
H 	Chain thin, thick at ends and changes of direction	H1 Cutting planes
J 	Chain thick	J1 Indication of lines or surfaces to which a special requirement applies
K 	Chain thin, double-dashed	K1 Outlines of adjacent parts K2 Alternative and extreme positions of movable parts K3 Centroidal lines

3. Geometric Construction:



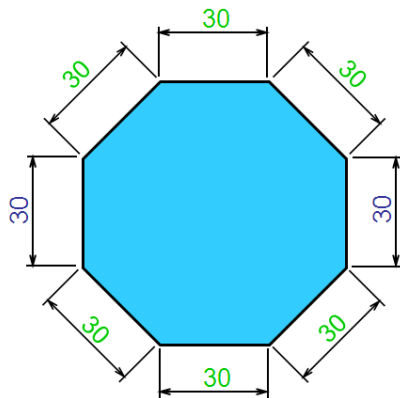
4. Dimensioning System:

General Principles:

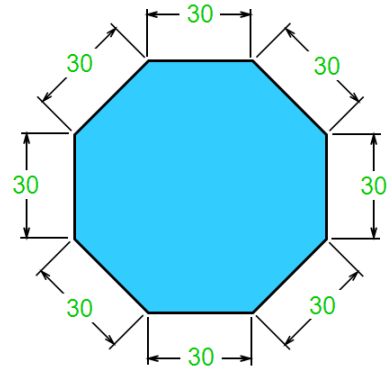
- ☐ All dimensions should be detailed on a drawing.
- ☐ No single dimension should be repeated except where unavoidable.
- ☐ Mark the dimensions outside the drawing as far as possible.
- ☐ Avoid dimensioning to hidden lines wherever possible.
- ☐ The longer dimensions should be placed outside all intermediate dimensions, so that dimension lines will not cross extension lines.

Elements of dimensioning:

1. Students should identify and know the correct drawing of the following dimensioning elements like Dimension lines, Extension lines, Leader lines, Arrowheads.
2. Draw the figure in both, Aligned system & unidirectional system.



Aligned System



Unidirectional System

Conversion of Units for Reference:

1 μ (1 micron)	=	0.000001 m (10^{-6} m)
1 mm (1 millimeter)	=	0.001 m (10^{-3} m)
1 cm (1 centimeter)	=	0.01 m (10^{-2} m)
1 dm (1 decimeter)	=	0.1 m (10^{-1} m)
1 dam (1 decameter)	=	10 m
1 hm (1 hectometer)	=	100 m (10^2 m)
1 km (1 kilometer)	=	1000 m (10^3 m)

Sheet 1: Orthographic Projections & Isometric View

BATCH A

1. Draw the following View for **Figure 1**

- a) Front View
- b) Top View
- c) Right hand side View

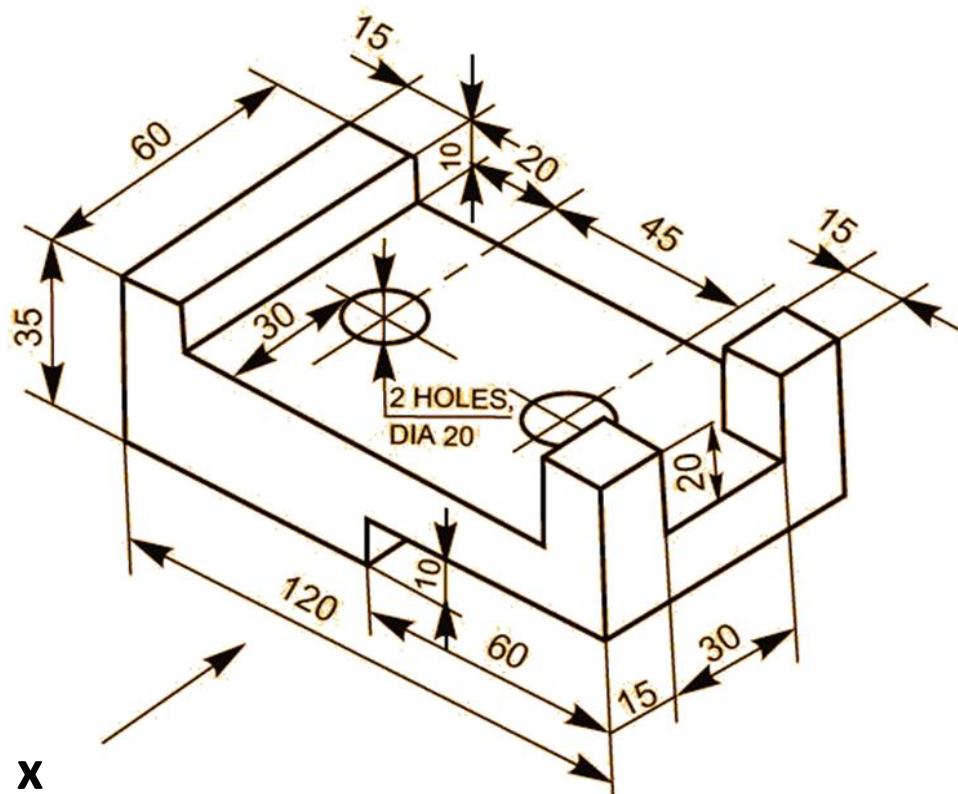


Figure 1

2. Draw the following View for **Figure 2**

- a) Right hand side view
- b) Top View
- c) Sectional FV

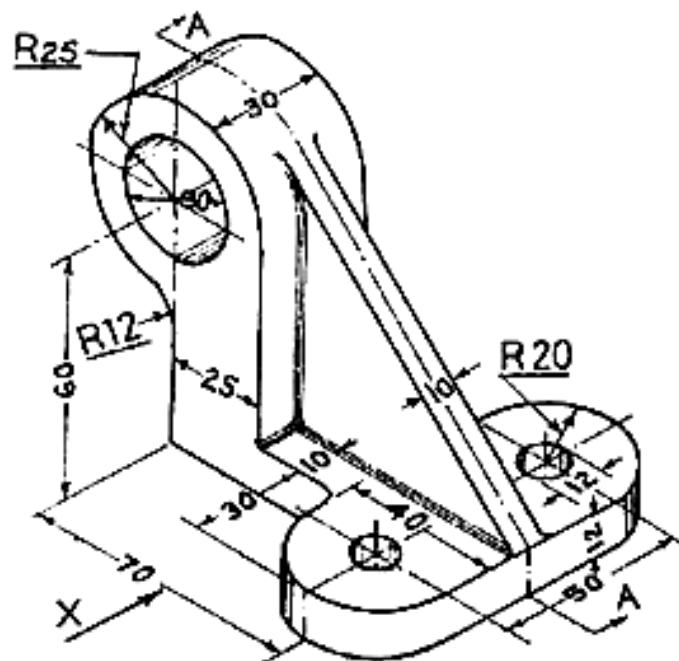
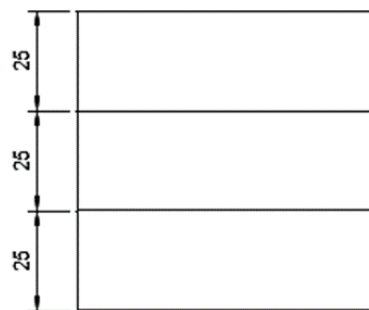
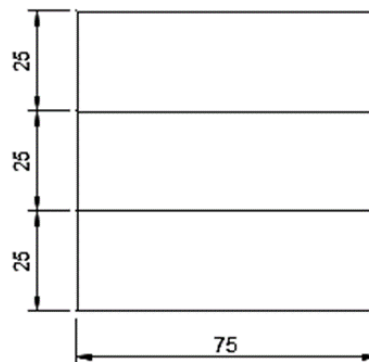


Figure 2

3. Draw Isometric View for **Figure 3**



FV



TV

Figure 3

BATCH B

1. Draw the following View for **Figure 4**

- a) Front View
- b) Top View
- c) Left hand side View

2. Draw the following View for **Figure 5**

- a) Front View
- b) Top View
- c) Sectional LHSV

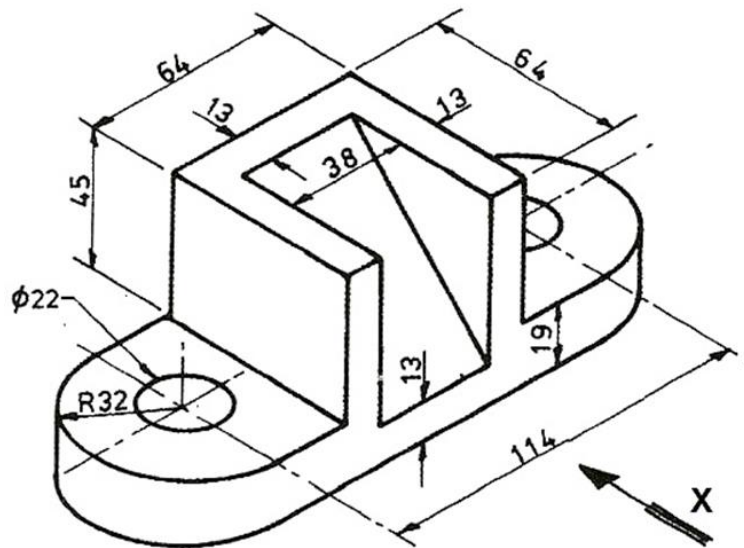


Figure 4

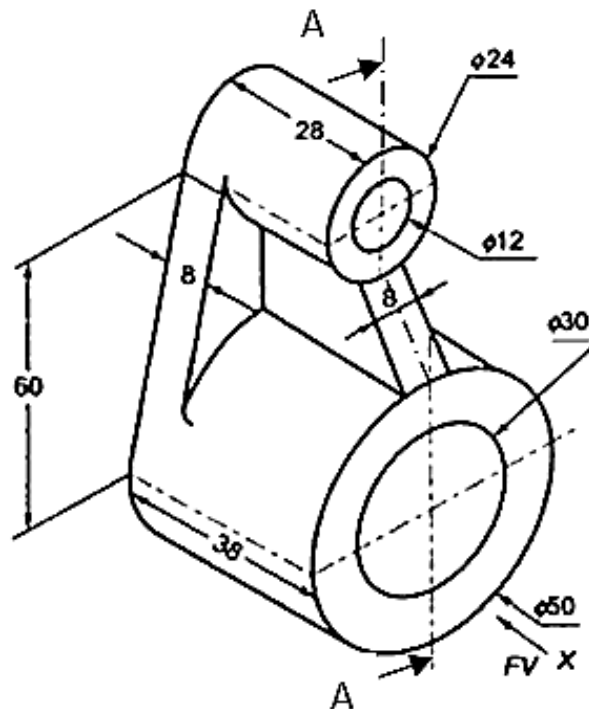


Figure 5

3. Draw Isometric View for **Figure 6**

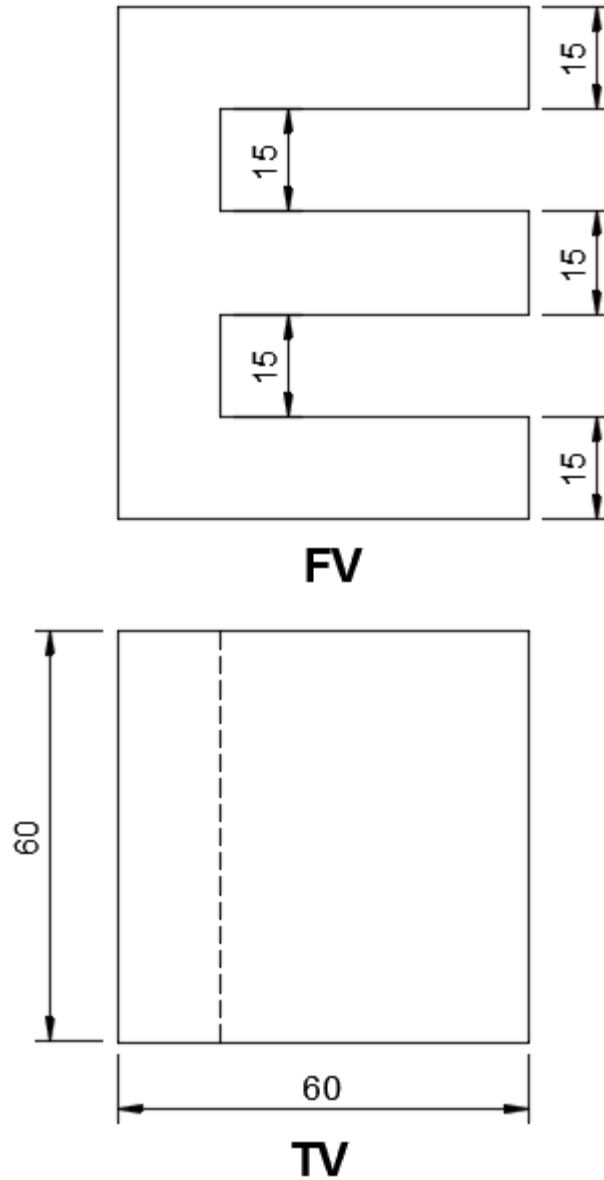
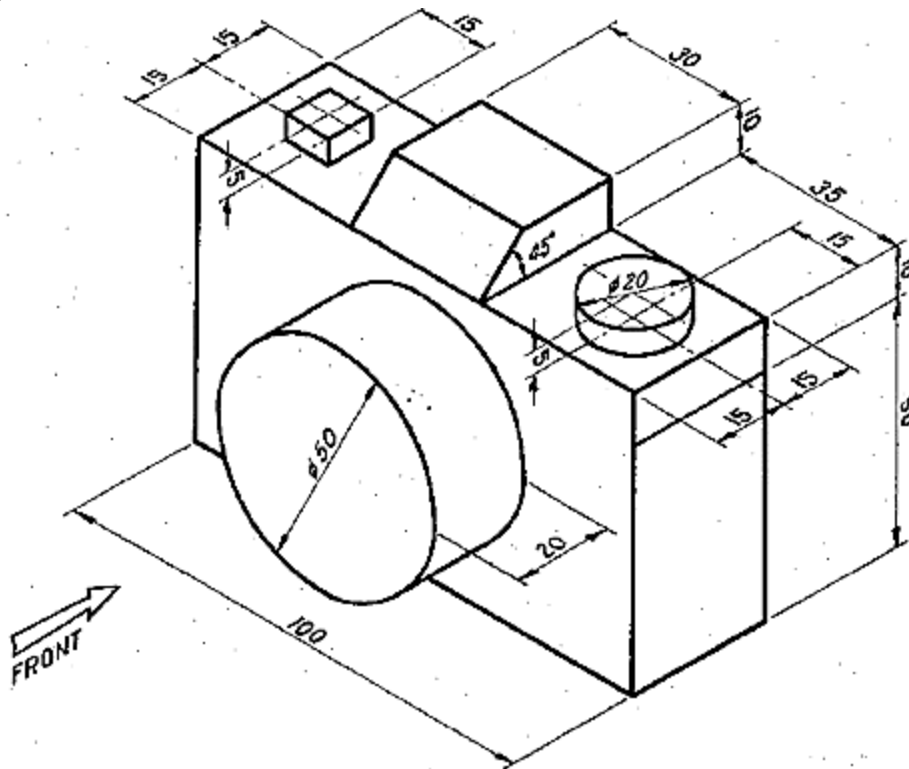


Figure 6

- Front View
- Top View
- Right hand side View



- Sectional Front View
- Top View
- LHSV

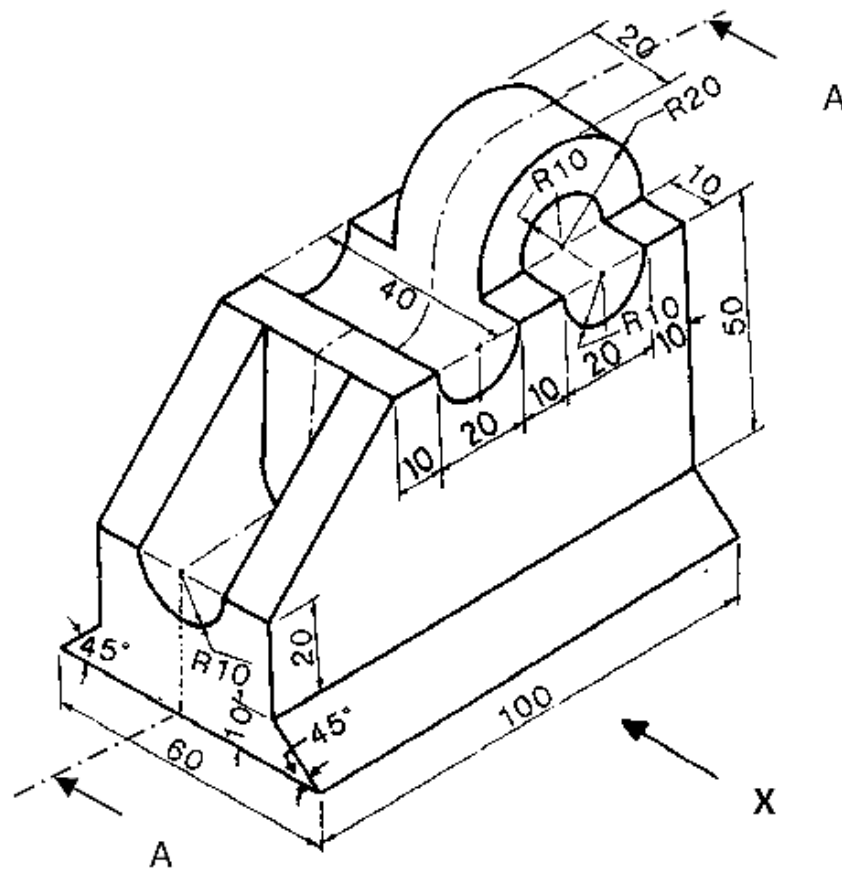


Figure 8

3. Draw Isometric View for **Figure 9**

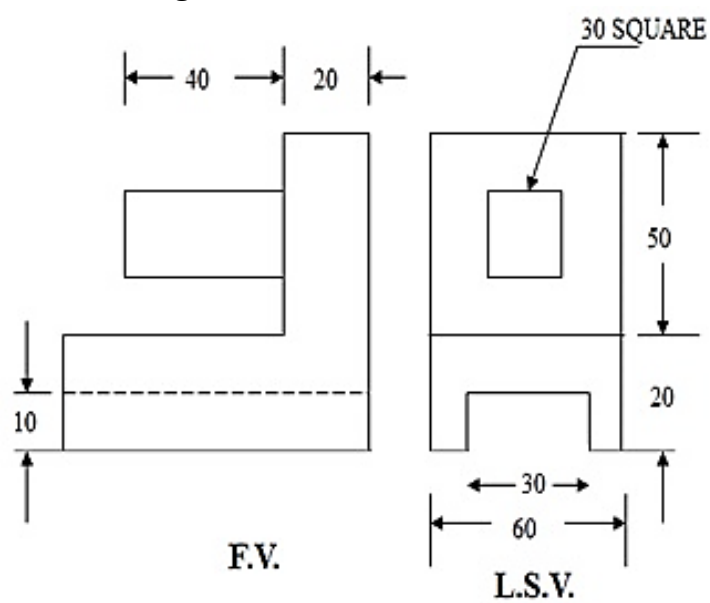


Figure 9

Sheet 2: Computer Aided Drafting and Modeling

BATCH A

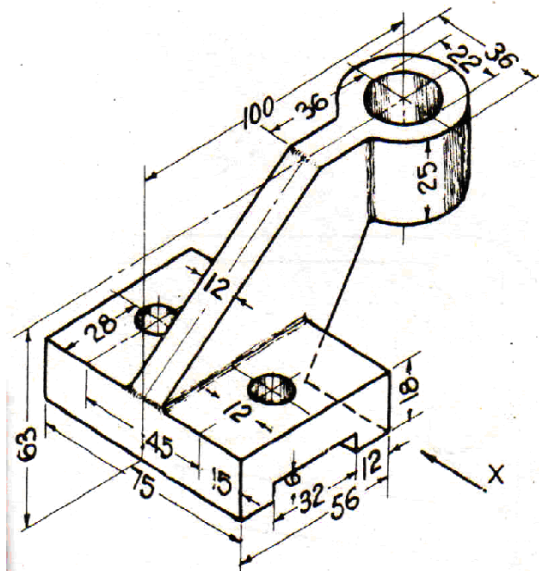


Figure 10

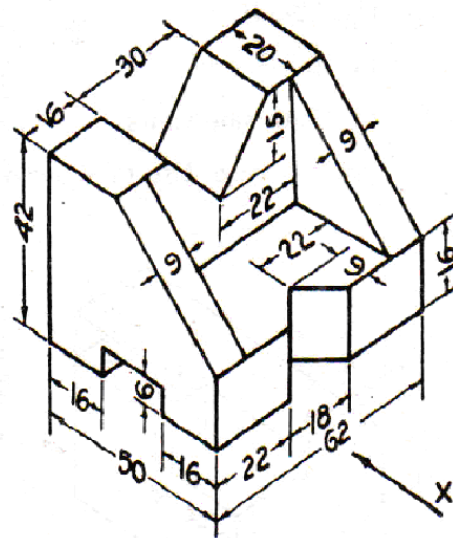


Figure 11

1. Draw the following View for Figure 10
 - a) Front View
 - b) Top View
 - c) Left hand side View
2. Draw the following View for Figure 11
 - a) Front View
 - b) Top View
 - c) Left hand side View

Note: Use any CAD software: Autocad /Creo /Autodesk fusion 360 to prepare the sheet.

BATCH B

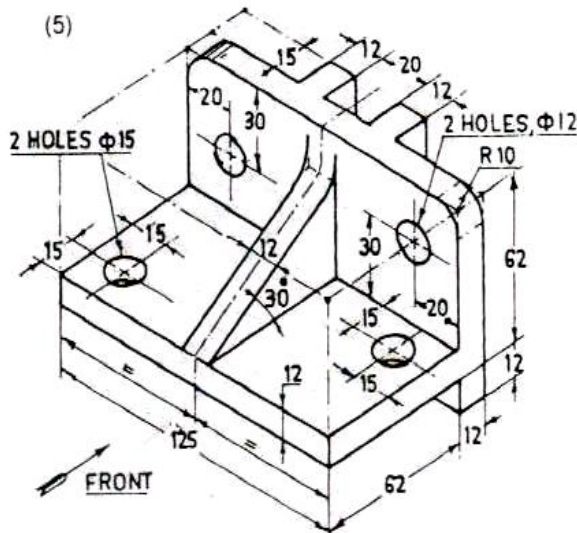


Figure 12

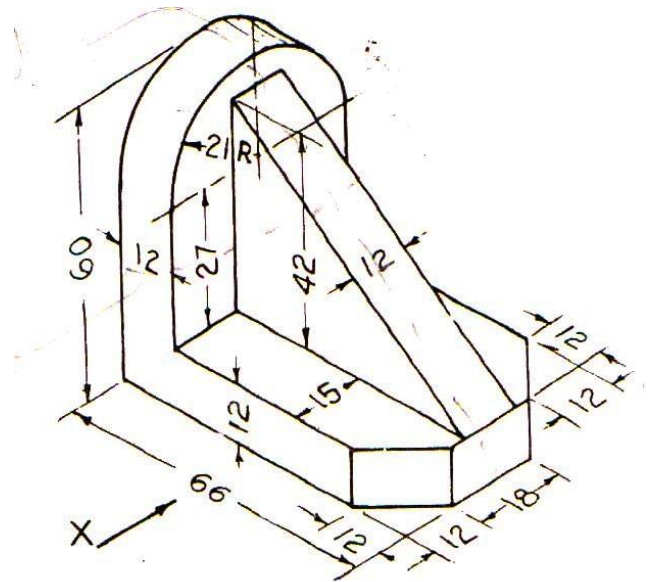


Figure 13

1. Draw the following View for Figure 12
 - a) Front View
 - b) Top View
 - c) Right hand side View
2. Draw the following View for Figure 13
 - a) Front View
 - b) Top View
 - c) Right hand side View

Note: Use any CAD software: Autocad /Creo /Autodesk fusion 360 to prepare the sheet.

Date:-

EXPERIMENT NO: 3

INTRODUCTION TO SURVEYING

OBJECTIVE:

To understand the fundamentals associated with Surveying.

THEORETICAL BACKGROUND:

Definition:

Surveying is an art of determining relative positions on, above or beneath the surface of the earth by measuring the horizontal distances & angles for determination of any point on ground.

Object of Surveying:

The object of surveying is to prepare a map / plan, with suitable scale, to show the relative positions of the objects on the surface of the earth.

- (i) Map
- (ii) Plan

Primary Divisions of Surveying:

- (i) Plane Surveying
- (ii) Geodetic Surveying

Principles of Surveying:

- (i) Working from whole to part not part to whole
- (ii) To locate the point by two reference points.

Classification of Surveying:

- A. Based on Nature of Field:
 - (i) Land
 - (ii) Hydrographic
 - (iii) Astronomical
 - (iv) Aerial
- B. Based on Object/ Purpose:
 - (i) Geological survey
 - (ii) Mine survey
 - (iii) Archeological survey
 - (iv) Military survey

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C. Based on Instruments:

- (i) Chain survey
- (ii) Compass survey
- (iii) Chain & Compass survey
- (iv) Plane Table survey
- (v) Theodolite Survey
- (vi) Tacheometry Survey
- (vii) Leveling Survey
- (viii) Photogrammetric survey
- (ix) EDM (Electromagnetic Distance Measurement) Survey

D. Based on Methods:

- (i) Triangulation
- (ii) Traversing

Units of Measure:

Units of Length

10 millimeters	=	1 centimeter
10 centimeters	=	1 decimeter
10 decimeters	=	1 meter
10 meters	=	1 decameter
10 decameters	=	1 hectometer
10 hectometer	=	1 kilometer
1 foot	=	12 inches
1 inch	=	2.54 cm
1 meter	=	3.28 feet
1 yard	=	3 feet
1 meter	=	1.093 yard
1 mile	=	1.6093 km

Units of Area

100 sq. millimeters	=	1 sq. centimeter
100 sq. centimeters	=	1 sq. decimeter

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100 sq. decimeters	=	1 sq. meter
100 sq. meters	=	1 sq. decameter or 1 are
100 sq. decameters or 1 are	=	1 sq. hectometer or 1 hectare
100 sq. hectometers	=	1 sq. kilometer
1 sq. mile	=	640 acres
1 acre	=	40 gunthas
1 acre	=	10 sq. chain (Gunters)
1 gunthas	=	1089 sq. ft

Units of Volume

1000 cu. Millimeters	=	1 cu. Centimeter
1000 cu. Centimeters	=	1 cu. Decimeter
1000 cu. Decimeters	=	1 cu. Meter
1 cu. Meter	=	219.969 gallons
1 cu. M	=	1 kilolitre
1 acre. Feet	=	271327 gallons

Exercise:

1. Explain role of civil engineers.
2. Define Surveying and state the objective of surveying.
3. Distinguish between plane survey and Geodetic survey.
4. What are the fundamental principles of surveying? Explain In briefly.

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Marks Obtained:	Signature of Faculty:	Date:
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Date:-

EXPERIMENT NO: 4

LINEAR MEASUREMENT-BY CHAIN SURVEY

OBJECTIVE:

To understand the method of linear measurements through chain survey.

To do Chaining, ranging and offsetting a survey line AB on a level ground.

APPARATUS:

Metric Chain - _____, Measuring Tape - _____, Wooden Pegs, Hammer, Ranging Rods, Steel Arrows.

THEORETICAL BACKGROUND:

Chain Surveying or Chain triangulation:

It is a method of surveying in which the area is divided into a network of triangles and the sides of the triangle are directly measured on the field with a chain or a tape and no angular measurements are taken. This type of surveying is used for smaller extent of area with simple details and ground is fairly level.

Terms related to Chain Surveying:

- i. Main station
- ii. Subsidiary or Tie station
- iii. Baseline
- iv. Survey line
- v. Check line
- vi. Tie line

Basic Operations in Chain Surveying:

(i) Chaining

The method of taking measurements with the help of a chain or a tape is termed as chaining.

(ii) Ranging

When the length of a line exceeds the length of the chain, to proceed in a straight line, intermediate points are needed to be established between the two stations. The process of establishing intermediate points on a straight line between two endpoints is known as ranging.

Methods of ranging

- (i) Direct Ranging
 - a. By Eye
 - b. By Line Ranger
- (ii) Indirect Ranging or Reciprocal Ranging

(iii) Offsetting

The method of locating (positioning) objects from the survey line by means of linear measurement (distance between survey line and object), right or left of the survey line is called as offsetting and lateral measurements (distances) are called offsets.

Types of Offsetting:

- (a) Perpendicular offset and (b) Oblique offset

INSTRUMENTS USED:

(A) For Chaining:

(i) Chains

1. Metric chain
2. Gunter's or Surveyor's chain
3. Engineer's chain
4. Revenue chain
5. Steel band

(ii) Tapes

1. Cloth or linen tape
2. Fiber tape
3. Metallic tape
4. Steel tape
5. Invar tape

(iii) Arrows

(iv) Peg

(v) Plumb bob

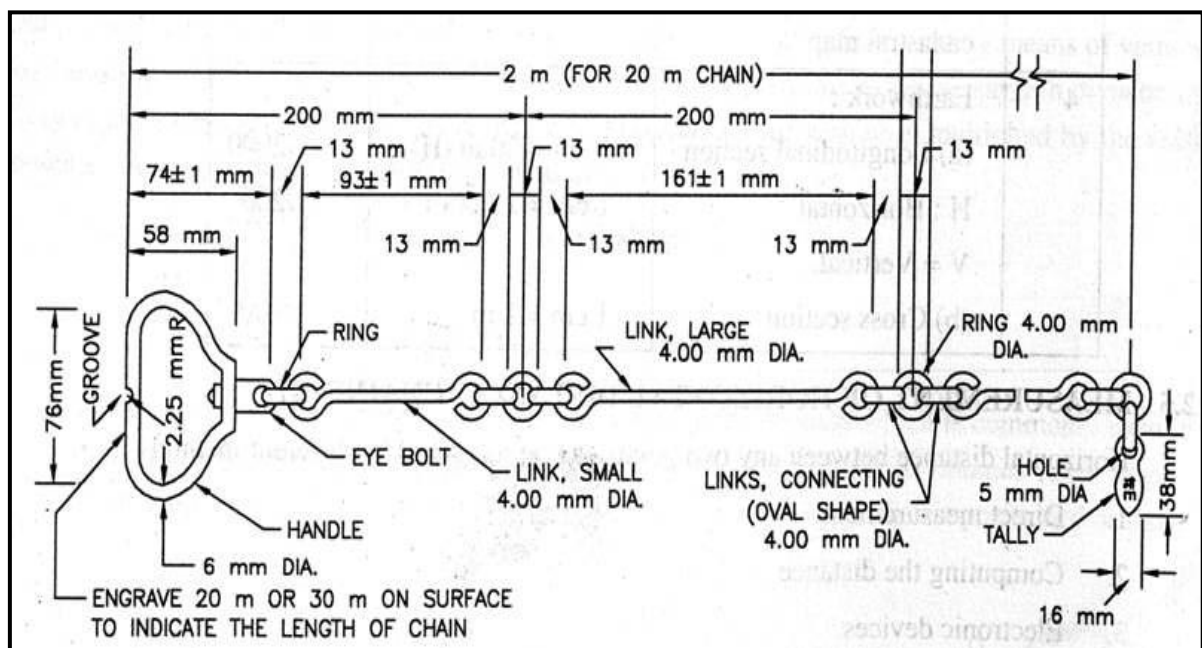


Figure: Details Of Metric Chain

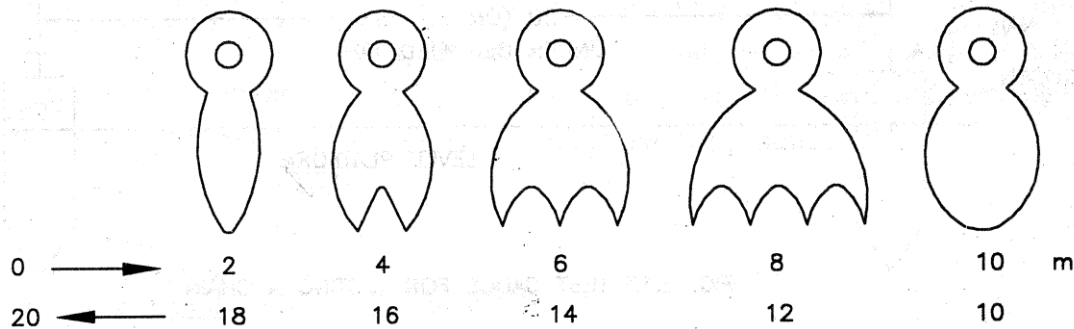


FIG. 2.12 TALLIES FOR A 20 METRE CHAIN

Figure: Tallies for a 20 m chain

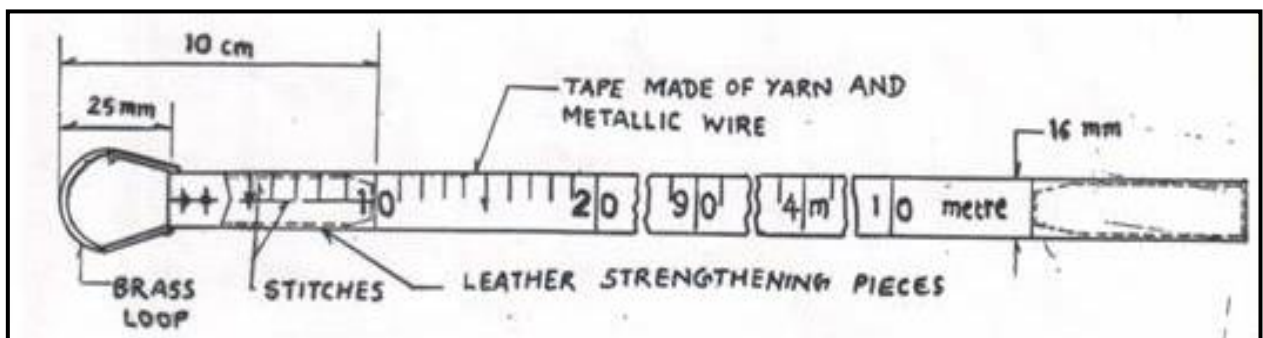


Figure : Measuring tape

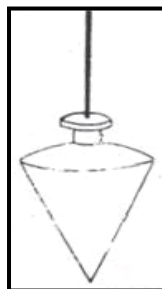
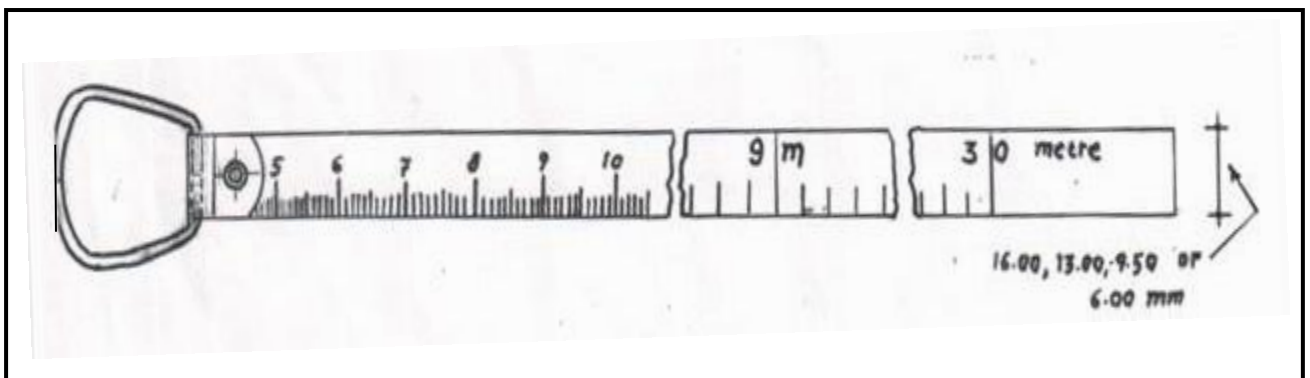


Figure :Plumb bob

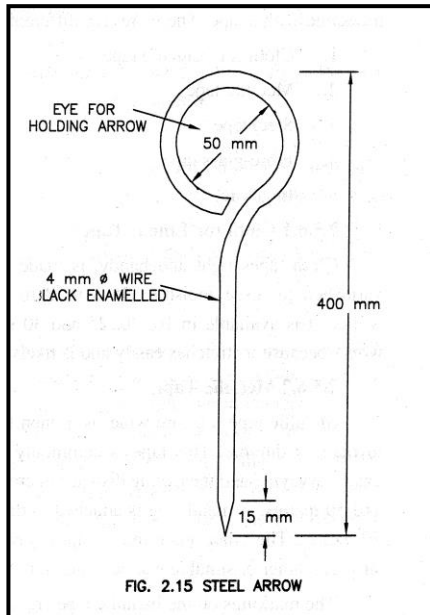


Figure : Steel Arrow

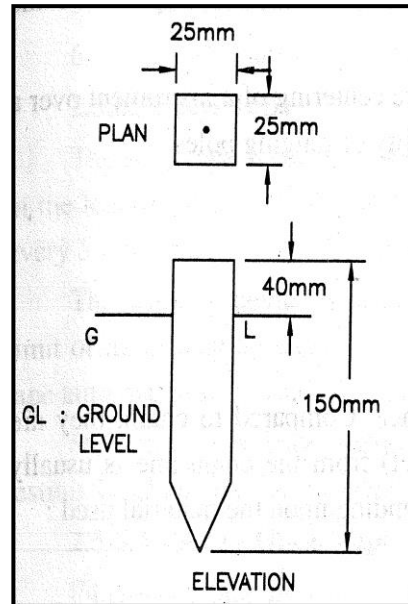


Figure : Wooden Peg

(B) For Ranging:

- (i) Ranging rod or Ranging Pole
- (ii) Offset rod
- (iii) Line ranger

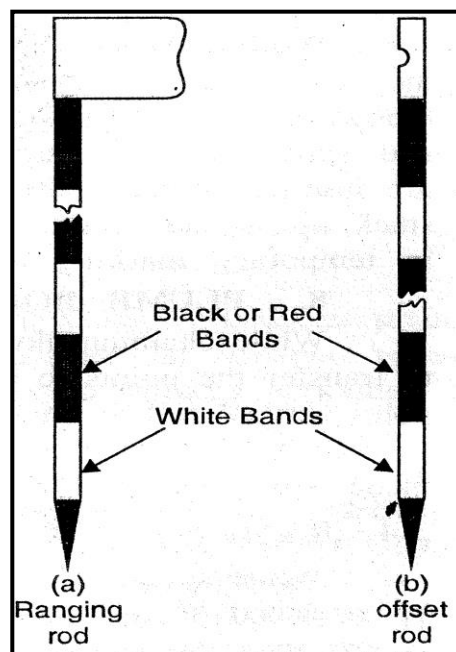


Figure : Ranging rod

(C) For Offsetting:

- (i) Cross staff
- (ii) Prism square
- (iii) Optical square

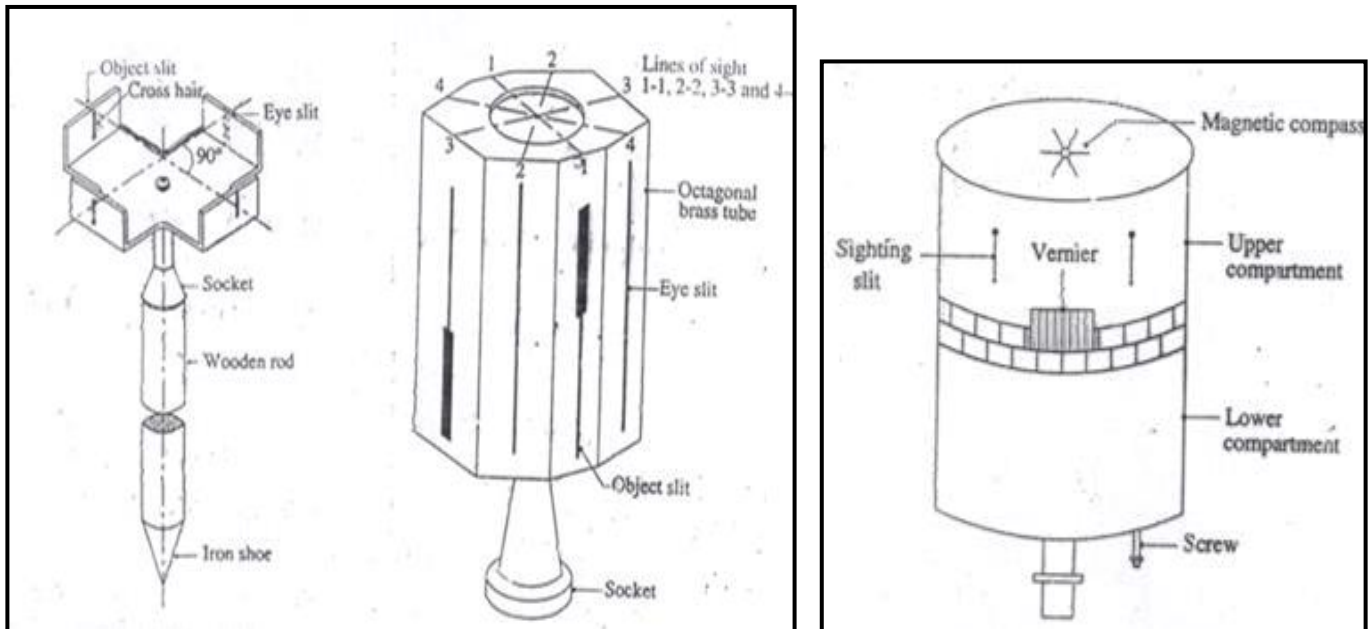


Figure :Cross staff

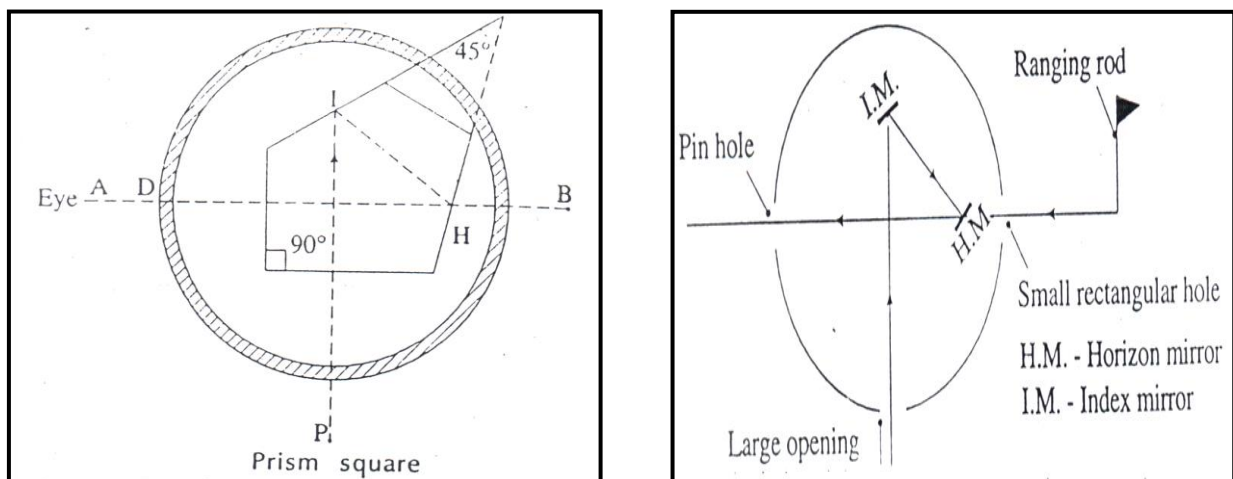


Figure :Prism square

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PROCEDURE:

1. Main stations A and B are first marked with pegs and ranging rods to make them visible.
2. Unfolding of chain is done and is laid between the station points to measure correct length of the line.
3. The chainman at the station A (At start of a line) is called as follower and who drag the chain forward along the line is called as leader. The follower places zero handle of the chain in contact with the peg at the beginning of the line and firmly holds it. The leader, taking ten arrows and a ranging rod in one hand and other handle of chain in other hand, walks off in forward direction.
4. If the length of the survey line is greater than one chain length, intermediate points are located, in order to pull the chain along a straight line. At the end of chain length, leader holds the rod vertically, approximately in line and then faces the follower. When intermediate ranging rod M is fixed on a straight line by direct observation, the process is known as direct ranging which is explained below.
5. The follower stands half a meter behind ranging rod at A by looking towards the line AB. The follower directs the leader to move ranging rod to the left or right until the three ranging rods at A, M and B come exactly at same straight line. The code of signals are used as below:

Sr. No.	Signals Given by the Follower	Meaning of the Signal
1	Rapid sweep with right hand	Move considerably to the right
2	Rapid sweep with left hand	Move considerably to the left
3	Slow sweep with right hand	Move slowly to the right
4	Slow sweep with left hand	Move slowly to the left
5	Right arm extended	Continue to move to left
6	Left arm extended	Continue to move to left
7	Right arm up and move to the right	Plumb the rod to the right
8	Right arm up and move to the left	Plumb the rod to the left
9	Both hands above head and brought down	Ranging is correct
10	Both arms extended forward horizontally and the hands brought down quickly.	Fix the ranging rod

6. The leader then marks the position of the rod by making a hole with it. Taking the handle in both hands and standing in line, leader straightens the chain by jerking and stretches it over the mark. Care should be taken that chain should be raised above the ground and should be straight as possible.
7. Leader inserts arrow against end of the chain in the ground, to mark the end of the chain. The leader then picks up other arrows, ranging rod, and drags the chain in forward direction. The leader goes on fixing arrows at the end of each and every chain length. After the tenth arrow is

- inserted (completing 10 m length), the follower collects all the arrows fixed by the leader and marks the end of the 10th chain by erecting a ranging rod.
8. At the end of the work, as a check the number of arrows inserted is multiplied by chain length which is total length measured.
 9. Fractional length of chain is found out by counting the links and multiplying it to size of each link. Fractional length is added to total length to find the true total length.
 10. Location of prominent objects can be located by offsetting on a chain line established above. The offsets are usually measured with tape.
 11. When offset is to be taken from chain line to the object, say P, the leader holds the zero end of tape at P and follower carrying tape box, swing it along the chain. Wherever the tape shows minimum length is considered to be foot of perpendicular (p), from point P at object, to chain line.
 12. Measure Ap and pP.
 13. Finally record the survey line AB and offsets in the field book.

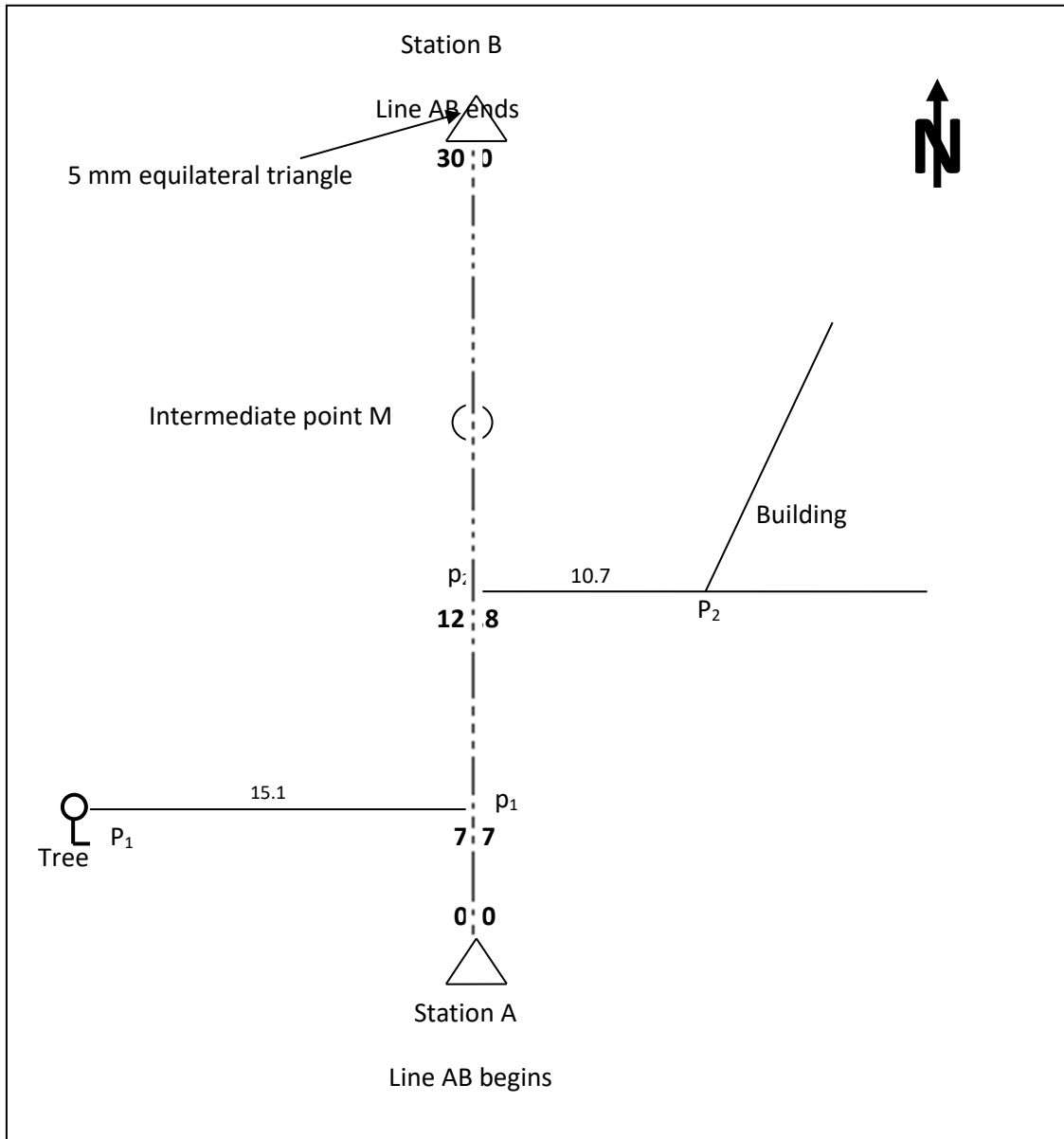


Figure : Representation of a field book entry page

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FIELD BOOK ENTRY PAGE



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CONVENTIONAL SYMBOLS

Sr. No.	Object	Symbol
1.	North Line	
2.	Main stations	
3.	Traverse stations or sub stations	
4.	Chain line	
5.	River	
6.	Canal	
7.	Open Well	
8.	Tube Well	
9.	Railway Line (single)	
10.	Railway line (Double)	
11.	Road Bridge or culvert	
12.	Railway Bridge or culvert	

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13.	Road & Rail level Crossing	
14.	Wall with gate	
15.	Building (Pakka)	
16.	Building (Katcha)	
17.	Temple	
18.	Bench Mark	
19.	Tree	
20.	Cultivated Road	
21.	Embankment	
22.	Cutting	
23.	Telephone Line	
24.	Telegraph Post	
25.	Electric Line	

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26.	Electric Post	
27.	Burial Ground or Cemetery	

Conclusion:-

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Date:-

EXPERIMENT NO: 5

CONSTRUCTION SITE VISIT

OBJECTIVE:

To visit a residential building / educational building and to study about various components of the building

THEORETICAL BACKGROUND:

Types of Building:

(A) Based on Occupancy as per the National Building Code:

- (i) Residential building
- (ii) Educational building
- (iii) Institutional building
- (iv) Assembly building
- (v) Business building
- (vi) Mercantile building
- (vii) Industrial building
- (viii) Storage building
- (ix) Hazardous building

(B) Based on Structure

- (i) Load bearing structure
- (ii) Framed structure
- (iii) Composite structure

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Components of Building:

Sr. No.	Components of Building	Functions
(A)	Sub-structure	
1	Foundation	
(B)	Super-structure	
1	Plinth	
2	Damp Proof Course	
3	Walls	
4	Floor	
5	Column	
6	Beam	
7	Opening in walls	
	(a) Doors	

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	(b) Windows	
	(c) Ventilators	
8	Sill	
9	Lintel	
10	Weather shed (Chhajja)	
11	Ceiling	
12	Roof	
13	Steps, Stair case & Lift	
14	Finishes for wall	
	(a) Plastering	
	(b)Painting/Varnishing/Polishing	
	(c) White washing	
15	Utility Fixtures	

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SITE VISIT REPORT

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EXPERIMENT NO: 6

BUILDING COMPONENTS DRAWING

OBJECTIVE:

To study the various components of a residential building.

THEORETICAL BACKGROUND:

A building has two basic parts:

- (i) Substructure or foundation and
- (ii) Superstructure.

Sub-structure or Foundation is the lower portion of the building, usually located below the ground level, which transmits the loads of the super-structure to the supporting soil. A foundation is therefore that part of the structure which is in direct contact with the ground to which the loads are transmitted.

The superstructure is that part of the structure which is above ground level, and which serves the purpose of its intended use. A part of the super-structure, located between the ground level and (the floor level is known as plinth.

The plinth is therefore defined as the portion of the structure between the surface of the surrounding ground and surface of the floor, immediately above the ground. The level of the floor is usually known as the plinth level. The built-up covered area measured at the floor level is known as a plinth area.

A building has the following components

1. Foundations
2. Masonry units : wall and columns
3. Floor structures
4. Roof structures
5. Doors, windows and other openings
6. Vertical transportation structures, such as stairs, lifts, ramps etc.
7. Building finishes.

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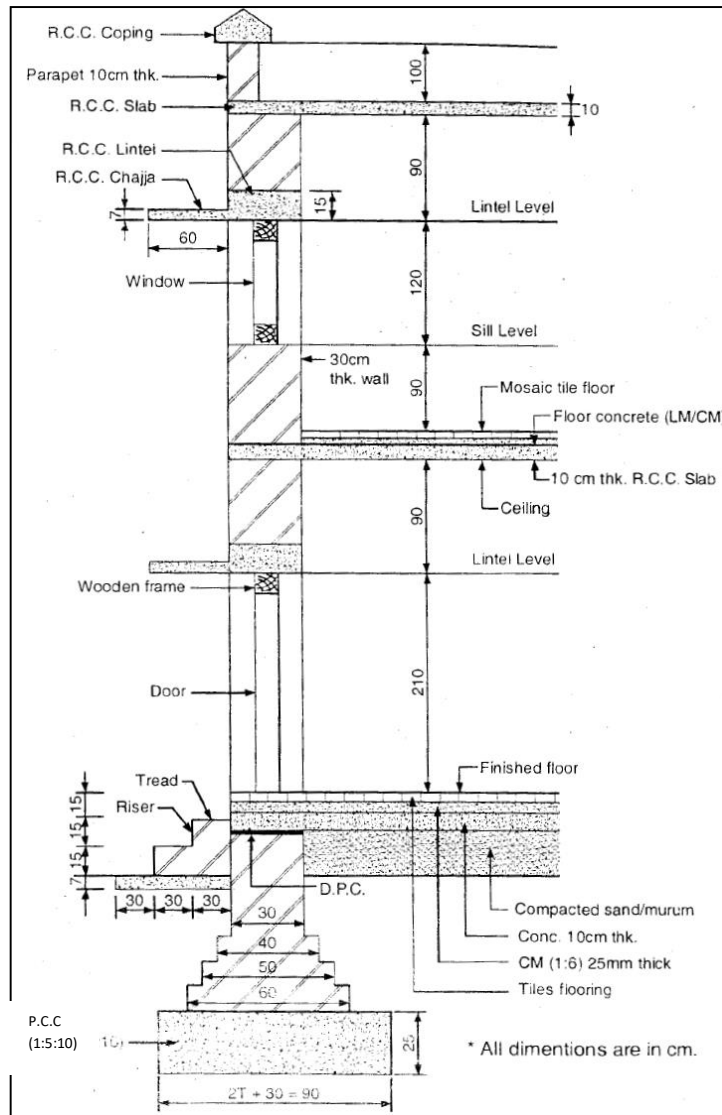


Figure: BUILDING COMPONENTS

(SECTION OF A 30 CM THICK MASONARY WALL)

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BUILDING COMPONENTS DRAWING:

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CROSS SECTION OF 20 CM THICK WALL FOOTING

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